

INDUSTRIAL TRAINING REPORT

STUDY OF INSTRUMENTATION, CONTROL AND COMMUNICATION SYSTEM IN A CEMENT MANUFACTURING PLANT

BACHELOR OF TECHNOLOGY
ELECTRONICS AND COMMUNICATION

SUBMITTED BY

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Training period: 24/06/2026 to 28/07/2026



SUBMITTED TO:

DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ABVGIT PRAGATINAGA, SHIMLA (H.P.)- 171202

JULY 2026

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30th July, 2026

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Rohit Singh**, a B. Tech student in **Electronics & Communication Engineering** from **Atal Bihari Vajpayee Government Institute of Engineering & Technology, Pragatinagar, Kothkhai, Shimla Himachal Pradesh**, successfully underwent **Industrial Training** at **UltraTech Cement Ltd., Baga Cement Works** from **24th June 2026 to 28th July 2026**.

During his training period, he gained exposure to **Instrumentation & Automation Systems**, including the fundamentals of **Programmable Logic Controllers (PLC)** and **Distributed Control Systems (DCS)**. He also developed a basic understanding of **Electrical & Instrumentation (E&I)** operations and their application in industrial processes.

Mr. Rohit Singh demonstrated a keen interest in learning, sincerity towards assigned tasks, and a disciplined approach throughout the training period. He actively participated in the learning process and exhibited good conduct, dedication, and professionalism during his tenure with the organization.

We wish him every success in his future academic pursuits and professional career.

For Ultratech Cement Limited
Unit: Baga Cement Works

Authorized Signatory



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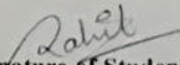


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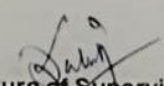
DECLARATION

I hereby declare that the Internship Report entitled ("Study of Instrumentation, Control and Communication System in Manufacturing Plant") is an authentic record of work carried out by me during my training at BAGA CEMENT WORKS, SOLAN from 24/06/2026 to 28/07/2026 under the supervision of my training supervisor Mr. Narendra Kumar (SH-E&I) And Mr. Ram Kumar Sharma (PF-E&I) for the award of degree of B.Tech. in Electronics & Communication Engineering.

Date: 28th July 2026


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I express my sincere gratitude to UltraTech Cement Ltd., Baga Cement Works, District Solan, Himachal Pradesh, for providing me the opportunity to undergo a 35-days industrial training at their esteemed organization. The exposure gained here has been invaluable in bridging the gap between theoretical knowledge acquired in the classroom and its real-world industrial application.

I am deeply thankful to the Management and Training Department of UltraTech Cement Ltd., Baga Cement Works, for granting permission and facilitating a smooth and productive training experience.

I extend my heartfelt thanks to all the engineers, supervisors and technical staff of the Instrumentation, Electrical and Process departments at Baga Cement Works, who generously shared their knowledge, guided me through the plant operations and answered my queries with patience and clarity throughout the training period.

I am grateful to the Department of Electronics and Communication Engineering, Atal Bihari Vajpayee Govt. Institute of Engineering and Technology (ABVG IET), Shimla, and the Head of Department for encouraging students to undertake industrial training as an integral part of the B.Tech. curriculum.

I also sincerely thank Himachal Pradesh Technical University (HPTU) for incorporating industrial training into the academic programme, thereby providing engineering students with practical exposure to real industrial environments.

Date: _____

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ORGANISATION PROFILE

Aditya Birla Group: A Premium Global Corporation

The Aditya Birla Group is a global conglomerate with a net worth exceeding \$65 billion USD, operating across metals, cement, textiles, chemicals, financial services, and telecommunications. Headquartered in Mumbai, India, the Group spans over 41 countries with a workforce of more than 140,000 employees, guided by its core values of integrity, passion, commitment, and seamlessness.

Baga Cement Works – Evolution & its growth

Baga Cement Works (BACW) is one of the latest technology cement plants, located at Tehsil Arki, District Solan, Himachal Pradesh. It became a UltraTech unit in 2017–18 following a change of ownership from Jaypee Associates Limited, effective 29.06.2017. The plant manufactures premium brand cement (43 & 53 grade) and fly ash blended cement using its own captive limestone mines. It has a designed clinker production capacity of 10,000 TPD, with an installed annual capacity of 33.0 lacs MT clinker and 17 lacs MT cement, feeding three grinding units at Bagheri (HP), Roorkee (UK), and Rajpura (PB).

Organization Environment

BACW follows a value-based participative management culture emphasizing people's development, innovation, and risk-taking abilities. The unit has won several awards including the Golden Peacock Environment Management Award, CII Quality Circle Debut Award, and FICCI Road Safety Award.

Vision, Mission & Values

Vision: To be the Leader in Building Solutions.

Mission: To deliver superior value to stakeholders on the four pillars of Sustainability, Customer Centricity, Innovation, and Team Improvement.

Values: Integrity, Commitment, Passion, Seamlessness, Speed.

Critical Success Factors:

Safety & Morale; increased use of Fly Ash and Bed Ash; maximizing clinker production; lowest manufacturing cost; best global energy efficiency; supplier partnership; use of alternate fuel; 100% blended cement production.

Product Offerings & Certifications

The unit follows a strong safety culture based on the DuPont methodology, applicable to all employees and contract workers. Key certifications and awards include IMS certification (ISO 9001, 14001, 45001, 50001, 27001), CII Energy Excellence Award, BIS Quality Excellence Award, and 5 Star Mines Award from IBM.

BACW produces Ordinary Portland Cement (OPC) and Portland Pozzolana Cement (PPC) under the national "UltraTech" brand, launched in 2018. Marketing is supported by a Technical Assistance and Services to Customer (TASC) wing, which liaises with plant experts to serve major project customers and build relationships with masons, builders, architects, and engineers. The company has focused market penetration across Himachal Pradesh, Punjab, and Jammu & Kashmir, with delivery ensured through EOW/SAP systems and real-time GPS monitoring for on-time, in-full (OTIF) delivery.

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CHAPTER 1

INTRODUCTION

Industrial training is an essential component of engineering education that provides students with an opportunity to apply theoretical knowledge in a real industrial environment. It bridges the gap between classroom concepts and practical engineering by exposing students to industrial processes, modern equipment, automation technologies, maintenance practices, and safety standards. For students of Electronics and Communication Engineering (ECE), industrial training offers valuable insight into the practical applications of sensors, instrumentation, industrial communication, control systems, and automation technologies that form the backbone of modern process industries.

The present industrial training was undertaken in the **Electrical and Instrumentation (E&I) Department** of **UltraTech Cement Ltd., Baga Cement Works**, Arki, District Solan, Himachal Pradesh. During the training period, practical exposure was gained to various process units, field instruments, automation systems, electrical equipment, and industrial safety practices employed in one of India's leading integrated cement manufacturing plants.

1.1 Background

1.1.1 Role of the Cement Industry in Infrastructure Development

The cement industry occupies a position of fundamental importance in the industrial and economic framework of any developing nation. As the principal binding material used in virtually all forms of construction, cement is indispensable to the creation of housing, commercial buildings, roads, bridges, dams, ports, and a wide range of industrial and public infrastructure. The pace at which a country develops its physical infrastructure is largely dependent on the availability, quality, and cost-effectiveness of cement produced within its borders. India today ranks among the largest producers of cement globally, and the sector continues to expand in response to sustained demand from housing programs, urbanization, and government-led infrastructure initiatives across the country.

UltraTech Cement Ltd. and Baga Cement Works

UltraTech Cement Ltd. is India's largest manufacturer of grey cement, ready-mix concrete, and white cement. The company operates several integrated cement plants across the country, utilizing advanced manufacturing technologies and automation systems to achieve high production efficiency and consistent product quality.

The **Baga Cement Works**, located at Arki in District Solan, Himachal Pradesh, is a fully integrated cement manufacturing plant equipped with modern process control systems, advanced field instrumentation, and centralized automation infrastructure. The plant utilizes Distributed Control Systems (DCS), Programmable Logic Controllers (PLC), industrial communication networks, and numerous field instruments to monitor and control the cement manufacturing process. These features make the plant an excellent learning environment for engineering students interested in industrial automation and instrumentation.

1.1.2 Economic and Regional Significance

Beyond its role in construction, the cement industry also contributes significantly to employment generation, ancillary industries such as mining, transportation, and packaging, and the overall economic activity of the regions in which cement plants are located. In hilly and resource-rich states such as Himachal Pradesh, cement manufacturing units play an especially important role, given the abundance of limestone reserves and the strategic importance of supplying cement to nearby northern markets including Punjab, Jammu and Kashmir, and neighbouring hill states.

1.1.3 Automation in Modern Cement Manufacturing

Modern cement manufacturing is no longer a manually operated, labour-intensive process. It has evolved into a highly automated, continuous, and technologically sophisticated process industry. From the extraction and crushing of raw materials to the blending, pyro processing in rotary kilns, and final grinding and packing of cement, every stage of production today is monitored and controlled through advanced instrumentation and automation systems. Distributed Control Systems (DCS), Programmable Logic Controllers (PLC), a wide array of field instruments, and industrial communication networks work together to ensure that the plant operates safely, efficiently, and with consistent product quality. This transformation of cement manufacturing into an automation-

driven process makes it an especially valuable environment for engineering students, particularly those from the Electronics and Communication Engineering (ECE) discipline to study and understand.

1.1.4 Industrial Training as an Academic Requirement

As part of the B.Tech curriculum in Electronics and Communication Engineering under Himachal Pradesh Technical University (HPTU), industrial training is a mandatory academic requirement prescribed for students, typically undertaken after the completion of the fourth semester. The purpose of this training is to expose students to real industrial environments, allowing them to observe first-hand how the theoretical concepts taught in the classroom — such as sensors, signal conditioning, control systems, and communication protocols — are actually implemented and applied at an industrial scale. It also provides an opportunity to understand professional work culture, safety practices, teamwork, and the practical challenges involved in operating and maintaining large-scale industrial systems. This structured exposure is considered essential in preparing engineering students for their eventual transition into professional careers, whether in industry, research, or further academic pursuits.

It is in this context that the present industrial training was undertaken at UltraTech Cement Ltd., Baga Cement Works, District Solan, Himachal Pradesh — a modern, fully integrated cement manufacturing unit known for its advanced automation infrastructure and commitment to operational excellence.

1.2 Objectives of the Training

1.2.1 Technical Objectives

- To gain practical, hands-on exposure to the industrial instrumentation and control systems employed in a large-scale, continuous process industry such as cement manufacturing.
- To understand the real-world application of core ECE concepts including sensors and transducers, signal conditioning circuits, closed-loop control systems, and industrial communication protocols, as they are actually implemented on the plant floor.
- To study, in detail, the architecture, configuration, and functioning of Distributed Control System (DCS) and Programmable Logic Controller (PLC) based automation systems used to monitor and control critical plant processes.

- To observe and understand the integration of field-level instruments such as temperature sensors, pressure transmitters, proximity switches, and gas analyser's with higher-level control and supervisory systems.

1.2.2 Academic and Professional Objectives

- To bridge the gap between theoretical knowledge acquired through academic coursework and its practical implementation in an operating industrial facility.
- To develop an appreciation for industrial safety practices, quality control procedures, and the overall organizational and operational culture of a large manufacturing enterprise.
- These objectives collectively aimed to provide a well-rounded understanding of how instrumentation and control engineering, a core area within the ECE discipline, is applied within a real, functioning process industry.

1.3 Scope of the Training

1.3.1 Duration and Location

The industrial training was carried out over a period of thirty days, commencing on 24th June 2026 and concluding on 28th July 2026. The training was hosted at UltraTech Cement Ltd., Baga Cement Works, located at Tehsil Arki, District Solan, Himachal Pradesh - a unit known for its advanced technological infrastructure and challenging mountainous operating environment.

1.3.2 Field Instrumentation Studied

Exposure was gained to various field instruments used throughout the plant, including resistance temperature detectors (RTDs) and thermocouples for temperature measurement, proximity switches, limit switches, pull-cord switches, and belt sway switches used for equipment protection and safety interlocking, as well as pressure transmitters, load cells, VFDs, and solenoid valves used in process actuation.

1.3.3 Control System Architecture Studied

Significant time was devoted to understanding the architecture and functioning of the plant's Distributed Control System, based on ABB's 800xA platform with AC 800M controllers, including the role of CI854 Profibus DP/V1 interface modules in facilitating communication between field devices and the control system.

1.3.4 Process Monitoring and Automation Systems

Additional areas of study included the monitoring systems employed for kiln operation, such as proximity-switch-based RPM detection and infrared shell scanning; automation associated with stacker-reclaimer systems; the working principles and integration of gas analysers used for emission monitoring and process control; and the broader distinction between PLC and DCS-based control architectures as applied within the plant.

A detailed and comprehensive account of these systems, their underlying working principles, and their role within the overall plant process is presented separately in Chapter III (Technical Content) of this report, which forms the core technical component of this submission.

1.4 Relevance of the Plant to the ECE Discipline

1.4.1 Instrumentation and Sensor Technology

Cement manufacturing plants rank among the most heavily instrumented and automated process industries in existence today, making them an exceptionally suitable environment for a student of Electronics and Communication Engineering to undertake industrial training. Nearly every stage of the manufacturing process from raw material handling to clinker production and final cement grinding is governed by an extensive network of sensors, transmitters, control loops, and communication systems. The wide use of field instruments and transducers throughout the plant offers direct, practical exposure to principles of sensors and instrumentation studied in the classroom.

1.4.2 Control Systems and Automation

The presence of sophisticated DCS and PLC-based automation systems within the plant directly parallels concepts covered in ECE coursework relating to control systems and industrial automation.

1.4.3 Industrial Communication Protocols

The industrial communication protocols used to link field devices with centralized control systems, such as Profibus, correspond closely to concepts of data communication and networking covered within the ECE curriculum. In this way, the plant environment provided a rich and relevant setting in which to observe the practical convergence of instrumentation, control systems, and communication engineering all core pillars of the ECE discipline.

1.5 Limitations of the Study

1.5.1 Constraints of Access and Confidentiality

While the industrial training provided substantial exposure to the plant's instrumentation and control systems, certain limitations were encountered during the course of the study. Access to some control room operations and process areas was restricted due to safety protocols and the proprietary nature of certain plant data, particularly relating to specific control logic configurations, process parameters, and production figures. As a result, some technical details presented in this report are based on general observation and discussions with plant personnel rather than direct hands-on operation of critical systems.

1.5.2 Time Constraints

The training period of thirty-five days, though substantial, was limited in scope when measured against the scale and complexity of a fully integrated cement manufacturing plant. Given the continuous nature of plant operations and the breadth of systems involved from raw material handling to final packing it was not possible to gain in-depth exposure to every subsystem within the available timeframe. Emphasis was therefore placed on areas most relevant to the ECE discipline, particularly instrumentation, control systems, and communication protocols.

1.5.3 Dependence on Plant Personnel and Documentation

A significant part of the learning during training relied on explanations provided by plant engineers and operators, as well as review of available technical documentation and manuals. While every effort was made to ensure the accuracy of the information presented in this report, some technical details are subject to the interpretation and understanding gained through these interactions and may not capture the complete depth of proprietary engineering design employed by the plant.

CHAPTER 2

WEEKLY PROGRESS REPORT

Training Duration: 24th June 2026 – 28th July 2026 (35 Days) | Organisation: UltraTech Cement Ltd., Baga Cement Works, District Solan, Himachal Pradesh

Table 2.1: Week 1 (24 – 30 June 2026): Plant Orientation & Field Instruments

Item	Details
Tasks Assigned	Plant induction, safety training; overview of cement manufacturing; study of field instruments.
Tasks Completed	Studied RTDs, thermocouples; proximity, limit, pull-cord, belt sway, and zero speed switches.
Plan for Next Week	Study pressure transmitters, load cells, solenoid valves; begin DCS introduction.
Problems/Remarks	Nonsignificant; plant-specific terms needed clarification from engineers.

Table 2.2: Week 2 (1 – 7 July 2026): Pressure, Actuation & DCS Introduction

Item	Details
Tasks Assigned	Study pressure transmitters, load cells, solenoid valves; introduction to ABB 800xA DCS.
Tasks Completed	Studied pressure transmitters, load cells, HYDAC EDS switches; began ABB 800xA / AC 800M study.
Plan for Next Week	Study CI854 Profibus modules; compare PLC vs. DCS architecture.
Problems/Remarks	Restricted access to some control room screens; explained verbally by supervisors.

Table 2.3: Week 3 (8 – 14 July 2026): DCS Architecture & Control Systems

Item	Details
Tasks Assigned	Study CI854 Profibus modules; PLC vs. DCS comparison; marshalling cabinets, motor control.
Tasks Completed	Studied Profibus DP/V1 modules, PLC/DCS comparison, XDI/XDO relay blocks, DOL starters.
Plan for Next Week	Study kiln monitoring systems and stacker-reclaimer automation.
Problems/Remarks	Nonsignificant.

Table 2.4: Week 4 (15 – 21 July 2026): Kiln & Material Handling Systems

Item	Details
Tasks Assigned	Study kiln monitoring and stacker-reclaimer automation systems.
Tasks Completed	Studied kiln RPM detection (2oo2 logic), IR shell scanner, stacker-reclaimer motions,
Plan for Next Week	Study VFDs, converter transformers, and gas analyzers.
Problems/Remarks	implementation ongoing.

Table 2.5: Week 5 (22 – 27 July 2026): Power Electronics, Gas Analysis & Report Compilation

Item	Details
Tasks Assigned	Study VFDs, converter transformers, gas analysers; consolidate learnings; compile report.
Tasks Completed	Studied VFD/IGBT switching, converter transformer protection, gas analyser integration (O ₂ /CO/NO _x). Consolidated learnings and compiled final report.
Plan for Next Week	Training concluded on 28th July 2026.
Problems/Remarks	None; overall training was informative with strong exposure to instrumentation and control systems.

CHAPTER 3

PLANT AND ITS PROCESS OVERVIEW

3.1 Overview of The Cement Manufacturing Process

Cement manufacturing is a continuous, energy-intensive process that transforms naturally occurring raw materials into a finely ground, chemically reactive binding material. The process, whether based on the wet, semi-dry, or dry method, fundamentally follows the same sequence of operations: extraction and preparation of raw materials, grinding and blending, preheating and calcination, clinkerization at high temperature, cooling, final grinding with additives, and packing. Modern plants, including Baga Cement Works, employ the dry process with pre-calciner technology, which is the most thermally efficient and widely adopted method used across the cement industry today. The overall sequence of operations, from raw material extraction to the dispatch of finished cement, is represented diagrammatically in Figure 3.1.

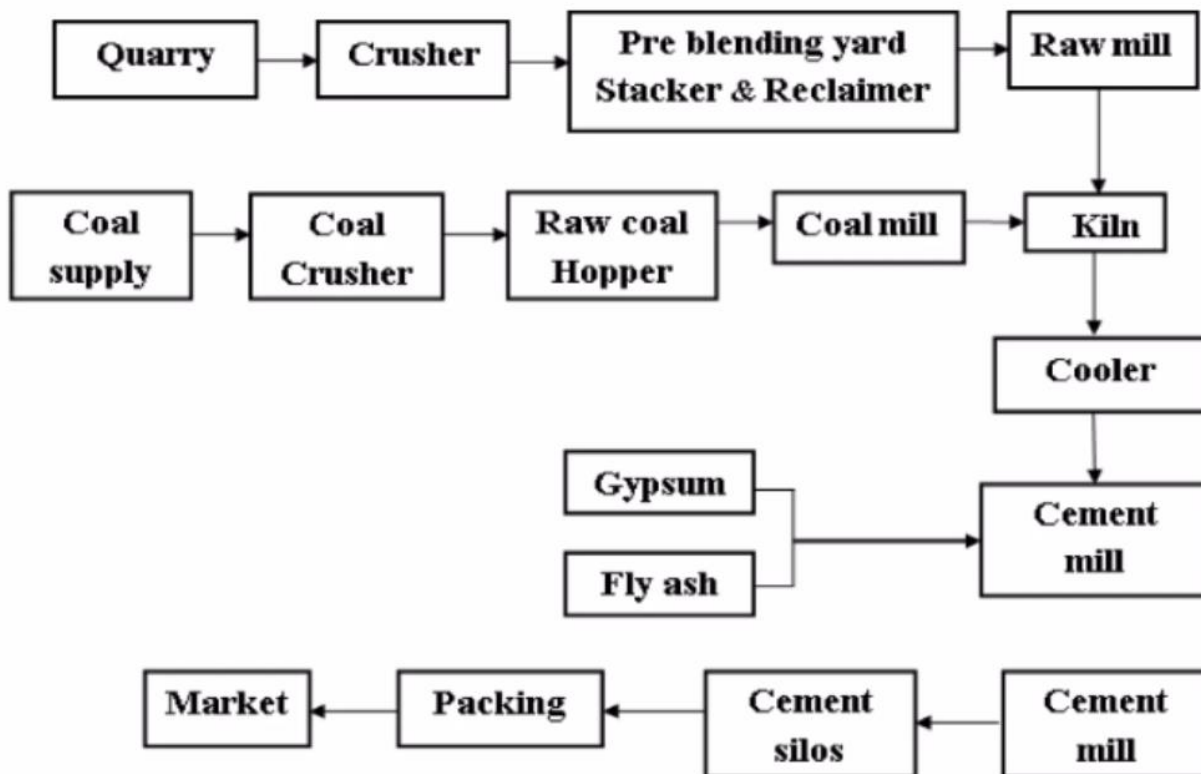


Fig 3.1 Process Flow Chart

Each of these stages involves a combination of mechanical processing, thermal treatment, and, increasingly, close monitoring through instrumentation and automation systems, making the overall process particularly relevant to the study of industrial control systems. The subsections that follow describe each stage of the process in detail.

Stage	Input	Output	Key Equipment
Quarrying	Limestone rock (mine)	Large limestone blocks	Excavator, dumper
Crushing	Large limestone blocks	Crushed limestone (<25 mm)	Jaw/Gyratory crusher, impact crusher
Stacking/pre-blending	Crushed limestone	Blended stockpile	Stacker-reclaimer
Proportioning	Limestone + clay + sand	Correct raw mix	Weigh feeder, load cells
Raw Grinding	Raw mix	Raw meal (fine powder)	Ball mill / VRM
Preheating	Raw meal	Pre-calcined meal	Cyclone preheater
Clinkerization	Pre-calcined meal	Clinker	Rotary kiln
Cooling	Hot clinker	Cooled clinker	Grate cooler
Cement Grinding	Clinker + gypsum	Cement powder	Ball mill / VRM
Dispatch	Cement powder	Bagged / bulk cement	Packing machine, trucks

Table 3.1: Summary of cement manufacturing stages

3.1.1 Quarrying and Raw Material Extraction

The cement manufacturing process begins with the quarrying of limestone, the primary raw material, from captive mines located in close proximity to the plant. Limestone provides the calcium carbonate required for the chemical reactions that follow in later stages. Along with limestone, supplementary raw materials such as clay, shale, and iron ore are also extracted or procured, as these provide additional silica, alumina, and iron oxide content necessary to achieve the correct chemical composition of the final clinker. The extracted limestone is typically blasted, excavated, and transported using heavy earth-moving machinery such as dumpers and loaders to the primary crusher located near the mine or plant boundary.

3.1.2 Crushing of Raw Materials

At the crusher, large lumps of limestone, often measuring up to a metre or more in size, are reduced to smaller fragments, generally in the range of 15–25 mm, suitable for further grinding. This is typically achieved using primary and secondary crushers, such as jaw crushers or hammer crushers, depending on the hardness and abrasiveness of the material. The crushed limestone is then conveyed via belt conveyors to the raw material storage yard or directly to the raw mill feed hoppers, where it awaits further processing.

3.1.3 Raw Meal Grinding and Blending

The crushed limestone, along with corrective raw materials such as clay and iron ore, is proportioned and fed into the raw mill. Within the raw mill, typically a vertical roller mill in modern plants, the material is ground into an extremely fine powder known as "raw meal." Precise proportioning at this stage is critical, as it determines the final chemical composition of the clinker and, consequently, the quality of the cement produced. The ground raw meal is then transported pneumatically to blending and storage silos, where it is homogenized to eliminate any compositional variation before being fed to the pyro processing system.

3.1.4 Preheating of Raw Meal

Before entering the kiln, the homogenized raw meal is fed into a multi-stage preheater tower, typically consisting of four to six cyclone stages arranged vertically. As the raw meal descends through the preheater, it comes into direct contact with hot exhaust gases rising from the kiln, in a counter-current arrangement. This process rapidly raises the temperature of the raw meal, driving off moisture and beginning the process of thermal decomposition,

while simultaneously recovering waste heat from the kiln exhaust gases, thereby improving the overall energy efficiency of the system.



Fig 3.2 Preheating and Cyclone Section

3.1.5 Precalcination

Following preheating, the raw meal passes through the precalciner, a specially designed combustion chamber positioned between the preheater tower and the kiln inlet. Here, a significant proportion of the calcination reaction — the decomposition of calcium carbonate (CaCO_3) into calcium oxide (CaO) and carbon dioxide (CO_2) — is completed before the material even enters the kiln. By achieving up to 90–95% calcination in the precalciner itself, the load on the rotary kiln is significantly reduced, allowing for higher production rates and improved fuel efficiency.

3.1.6 Clinkerization in the Rotary Kiln

The precalcined material then enters the rotary kiln, a large, slightly inclined, rotating cylindrical vessel lined internally with refractory bricks. As the material moves through the kiln under gravity and rotational motion, it is progressively heated by a counter current flow of hot combustion gases generated by the burning of fuel (coal, pet coke, or alternate fuels) at the kiln's discharge end. Temperatures within the burning zone reach approximately

1400–1450°C, at which point a series of complex chemical reactions occur, resulting in the formation of clinker — small, dark, glassy nodules composed primarily of calcium silicates, aluminates, and ferrites, which constitute the core reactive component of cement.



Fig 3.3 Clinkerization, Rotary Kiln, Cooling and Coal Dosing Section

3.1.7 Clinker Cooling

Upon discharge from the kiln, the red-hot clinker, at temperatures of around 1200°C, is rapidly cooled to approximately 100–200°C in a clinker cooler, typically of the grate or planetary type. Cooling is achieved through the forced passage of ambient air across the clinker bed. This rapid cooling is essential, as it locks in the desired mineralogical phases within the clinker, preventing the formation of undesirable crystalline structures that could adversely affect cement quality. Additionally, the hot air recovered from the cooler is recirculated back into the kiln and precalciner as secondary and tertiary combustion air, further improving the thermal efficiency of the overall system

3.1.8 Clinker Storage

The cooled clinker is transported via conveyors to clinker storage silos or domes, where it is stockpiled prior to final grinding. This storage stage also provides a buffer, allowing continuous kiln operation even during short-term fluctuations in downstream grinding or dispatch operations.

3.1.9 Cement Grinding

The stored clinker, along with gypsum (added to regulate the setting time of cement) and other additives — such as fly ash, slag, or limestone, depending on the specific grade or type of cement being manufactured — is fed into the cement mill. Within the cement mill, again typically a vertical roller mill or a ball mill in older or supplementary installations, the material is ground to the required fineness, resulting in the final product: cement. The proportioning of additives at this stage determines whether the final product is Ordinary Portland Cement (OPC), Portland Pozzolana Cement (PPC), or another blended variant.

3.1.10 Cement Storage, Packing, and Dispatch

The finished cement is conveyed to cement storage silos, segregated by grade and type. From the silos, cement is drawn for final packing, either into standard 50 kg bags using automated rotary packing machines or loaded directly in bulk form into specially designed bulk tankers for large-scale customers. Dispatch is carried out via road transport, supported by systems such as SAP and Electronic Order and Weighing (EOW) mechanisms to ensure accurate, timely, and traceable delivery to customers across the plant's primary markets.



Fig 3.4 Packing and Dispatch Section

3.2 Raw Material Handling — Quarrying, Crushing and the Stacker-Reclaimer System

3.2.1 Quarrying

Limestone is the primary raw material for cement, providing the calcium carbonate (CaCO_3) required for clinker formation. At Baga Cement Works, limestone is quarried from an open-cast mine in close proximity to the plant. The quarrying process involves drilling holes in the rock face, filling them with explosives and blasting to break the rock into large fragments. Excavators and front-end loaders then load the broken rock onto dumper trucks, which transport it to the crusher.

3.2.2 Crushing

The crushed limestone arrives at the crusher house in large lumps that are too big to be fed directly into the raw mill. The crushing stage therefore reduces the material to a size suitable for grinding. At this plant, two stages of crushing are employed:

- Primary Crusher: Receives large limestone rocks directly from the quarry dumpers. A jaw or gyratory crusher is typically used here, reducing material to a coarser size. The crushing action is achieved by compressing the rock between a fixed plate and a moving plate.
- Secondary Crusher: Receives the coarser output of the primary crusher and further reduces it to a finer fraction (typically below 25 mm) through impact. Hammer or impact crushers are commonly used at this stage.

The material discharged from the secondary crusher is then carried by belt conveyors to the raw material stockyard for stacking.

3.2.3 The Stacker

A stacker is a large mechanical conveying machine whose function is to deposit the crushed limestone onto a long, organised stockpile in the raw material yard. The stacker at this plant is a combined type, incorporating three independent mechanical motions:

- Travelling: The entire stacker machine moves forward and backward along a rail track installed in the stockyard. This longitudinal movement allows the stacker to build a long pile rather than depositing all material in one spot.
- Slewing: The boom conveyor (the long arm of the stacker that carries and discharges material) can rotate horizontally to the left and right. This lateral motion allows material to be spread across the width of the pile.
- Luffing: The boom can also be raised or lowered (like tilting an arm up or down). This vertical adjustment controls the height from which material is dropped onto the pile. Keeping the drop height small reduces dust generation and minimises material degradation due to impact.

Material is fed onto the stacker's own boom conveyor from the yard belt conveyor through a transfer chute. As the boom conveyor carries the material to the tip of the boom, the combined travel, slew and luff motions deposit it in thin, overlapping layers across the full length and width of the pile.



Fig. 3.5 Stacker and Reclaimer

3.2.4 Stacking Pattern and Pre-blending

The stacker builds the stockpile following the Chevron stacking pattern: the stacker travels from one end of the pile to the other, depositing a thin layer each pass, then returns and deposits another layer on top. This is repeated until the desired pile height and volume are achieved.

The reason for building the pile in layers is pre-blending, also called homogenization. Raw limestone from the quarry naturally varies in its chemical composition (calcium carbonate content, silica content, etc.) from one blast to another or from one part of the quarry to another. By depositing successive layers of material with slightly different compositions on top of each other, the variations are distributed throughout the height of the pile. When the reclaimer later retrieves material by cutting vertically through the pile, it picks up a sample from every layer simultaneously, so the output is an averaged, consistent composition. This consistency is critical for maintaining the correct clinker chemistry downstream.

Stacking Pattern	Description	Blending Quality
Chevron	Stacker travels full pile length repeatedly, building successive layers	Good — most common in cement plants
Windrow	Multiple parallel chevron piles-built side by side, reclaimed together	Excellent — used where highest blending is needed
Chevcon	Combination of chevron and conical patterns	Moderate

Table 3.2: Stacking patterns and blending quality

3.2.5 Mode of Operation — Manual and Automatic

The stacker can be operated in two modes, selectable via a Manual/Auto selector switch on the local control panel:

- Manual Mode: The operator directly controls the travel, slew and luff motions individually using joystick controls or push-buttons on the local panel. The operator visually monitors the pile and adjusts the movements accordingly. Safety interlocks (limit switches at the rail-track ends) remain active even in manual mode and will stop the machine if the travel limit is reached.
- Automatic Mode: A pre-programmed PLC sequence controls all three motions automatically, following the programmed stacking pattern. The stacker travels back and forth, slewing and luffing as required without operator intervention. The operator monitors the process from the control room SCADA screen and can intervene by switching to manual if required.

3.2.6 The Reclaimer

The reclaimer is the machine that retrieves the pre-blended material from the stockpile and feeds it forward to the proportioning equipment. It operates in the same rail track as the stacker (though not simultaneously). A bucket-wheel reclaimer uses a rotating wheel fitted with buckets to scrape material from the face of the pile as the machine travels along the rail, dropping it onto an underlying belt conveyor that carries the material out of the yard. Limit switches at the ends of the rail track also protect the reclaimer from overtravel.

3.3 Bucket Elevators and Material Conveying

3.3.1 Need for Vertical Transport

Throughout the cement plant, material must be moved not only horizontally (by belt conveyors) but also vertically, often by considerable heights. For example, raw meal must be lifted from the raw mill discharge (at ground level) to the top of the preheater tower (which may be 80–100 m tall). Ground clinker and cement must be lifted to the tops of their respective silos for gravity-assisted filling. For these vertical transport duties, bucket elevators are used.

3.3.2 Construction and Working Principle

A bucket elevator consists of a series of metal buckets attached at regular intervals to a continuous chain or belt running between two sprockets or pulleys — one at the top (head) and one at the bottom (boot). The entire assembly is enclosed in a tall, narrow casing.

Working principle: Material is fed into the casing at the bottom (boot) section. As the chain/belt moves upward, the buckets scoop up the material. When the buckets reach the head sprocket and travel over it, the centrifugal force (in centrifugal-discharge type) or gravity (in continuous-discharge type) causes the material to be thrown out of the buckets and into the discharge chute, which directs it to the next stage of the process.

Table 3.3: Types of bucket elevators and their applications

Type	Discharge Method	Best Suited For
Chain bucket elevator (central chain)	Centrifugal or gravity	Heavy, abrasive material — clinker, raw limestone
Belt bucket elevator	Centrifugal	Lighter material — raw meal, cement powder
Centrifugal discharge type	Centrifugal force at head	Fine, free-flowing material at high speed
Continuous (gravity) discharge	Gravity between closely spaced buckets	Lumpy or fragile material requiring gentle handling

3.3.3 Applications in the Cement Plant

Bucket elevators are found at multiple points in the process flow:

- Raw meal elevator: Lifts ground raw meal from the raw mill discharge to the top of the cyclone preheater tower — typically the tallest elevator in the plant.
- Clinker elevator: Carries cooled clinker from cooler discharge to clinker silo / cement mill feed hopper.
- Cement elevator: Lifts finished cement from the cement mill to the cement silos.

3.3.4 Belt Conveyors

Belt conveyors are used for all horizontal and moderately inclined material transport in the plant — from the crusher to the stacker, from the reclaimer to the proportioning bins, and between most other process stages. They consist of a continuous rubber belt supported by idler rollers and driven by a motorized head pulley. Key safety devices fitted on every belt conveyor include:

- **Pull-cord (pull-rope) switches:** A rope running the full length of the conveyor, connected at intervals to limit switches. If any person pulls the rope in an emergency, the circuit is broken and the conveyor stops immediately.
- **Belt-sway switches:** Detect if the belt has moved sideways off-centre (a common fault caused by uneven loading or misaligned idlers). If sway exceeds a set limit, the switch trips the conveyor to prevent belt damage and material spillage.
- **Zero-speed switches (proximity type):** Monitor the rotation of the tail pulley. If the belt stops while the motor is still running (indicating belt slip or breakage), the proximity switch detects the absence of pulse signals and triggers a trip.



Fig.3.6 Bucket Elevator and Conveyor

CHAPTER 4

POWER ELECTRONICS

4.1 Motor Starters

Large induction motors driving conveyors, fans, mills and the kiln cannot simply be switched directly onto the supply voltage in all cases. When a motor is started by connecting it directly to full voltage (DOL starting), it draws a large inrush current (typically 5–7 times the full-load current) from the supply. For very large motors, this inrush current can cause voltage dips in the plant electrical distribution system and mechanical shock to the driven equipment. Different starting methods are used to limit this effect:

Starter Type	Principle	Advantages	Application in plant
DOL (Direct On Line)	Motor connected directly to full voltage at start	Simple, low cost, full torque immediately	Small motors (<7.5 kW) — pumps, small conveyors
Star-Delta	Motor starts in star connection (reduced voltage), then switches to delta (full voltage)	Reduces starting current to about 1/3 of DOL current	Medium motors (7.5–75 kW) — fans, medium conveyors
Soft Starter	Electronic thyristor circuit gradually ramps up voltage from zero to full	Smooth start, no mechanical shock, adjustable ramp time	Pumps, fans, conveyors where smooth starting is important
VFD (Variable Frequency Drive)	Full variable speed control via AC–DC–AC conversion	Precise speed control, maximum energy saving, very soft start	Kiln drive, raw mill fan, cement mill separator, ID fan

Table 4.1: Motor starter types and applications

4.2 Direct-On-Line (DOL) Starter

A Direct-On-Line starter is the simplest form of motor starting method, in which full line voltage is applied directly to the motor terminals at the instant of starting. It is widely used in cement plants for smaller induction motors (typically up to 15–20 kW) driving auxiliary equipment such as conveyor idlers, small fans, lubrication pumps, and agitators, where high inrush current at starting is not a concern for the supply network.

Construction and Components

A DOL starter panel, housed within the Motor Control Centre (MCC), typically comprises:

Isolator/MCCB — for manual isolation and short-circuit protection

Contactor — an electromagnetically operated switch that connects the motor to the supply when energized by the control circuit

Overload relay (thermal or electronic) — protects the motor against sustained overcurrent due to mechanical jamming, single-phasing, or overloading

Control circuit — comprising Start/Stop push buttons, auxiliary contacts for interlocking, and indication lamps.

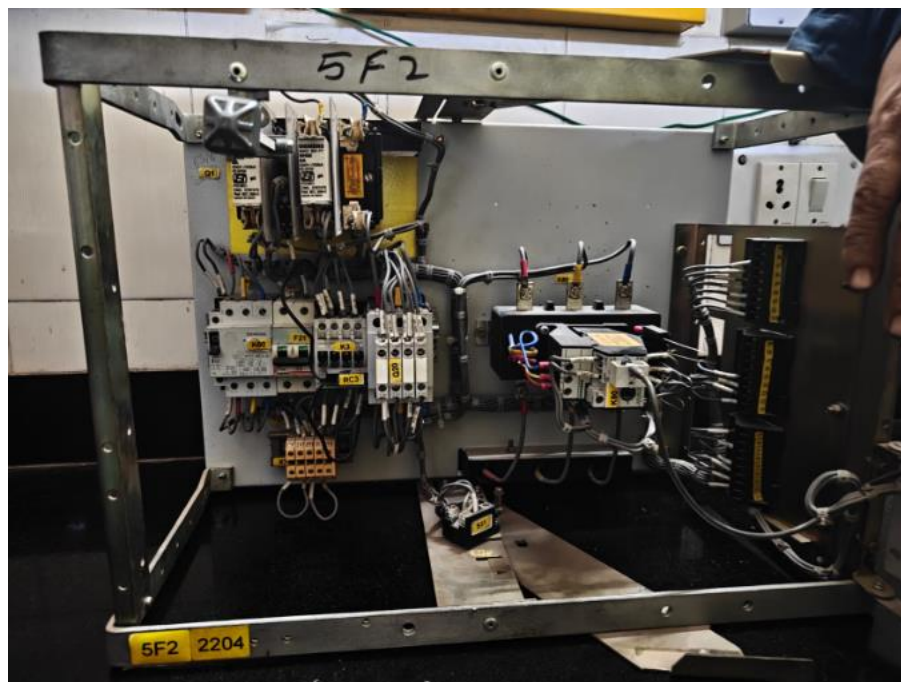


Fig 4.1 DOL Feeder

Working Principle:

On pressing the Start push button, the contactor coil is energized, closing its main contacts and applying full three-phase supply directly to the motor stator windings. The motor accelerates from standstill to full speed while drawing a starting current of roughly 5–7 times its rated full-load current. A normally-open auxiliary contact on the contactor is wired in parallel with the Start button to maintain coil energization after the button is released (seal-in/holding circuit). Pressing Stop, or the operation of the overload relay on a fault, de-energizes the coil and disconnects the motor.

Role of the Overload Relay:

The overload relay continuously monitors motor current through current transformers or bimetallic elements. If current exceeds the set threshold for a defined time (matching the motor's thermal withstand curve), it trips its contact, breaking the contactor coil circuit and stopping the motor — protecting the windings from thermal damage.

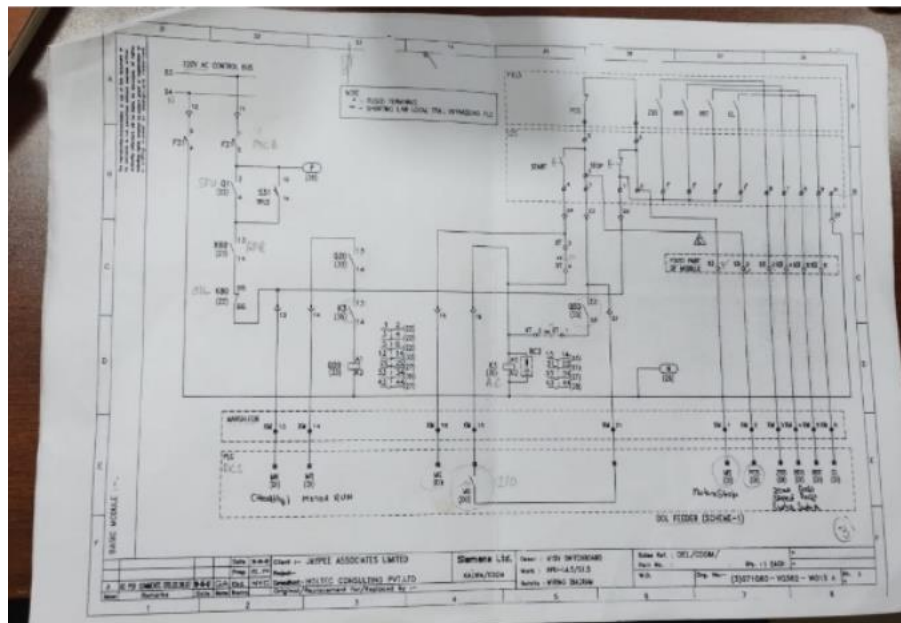


fig 4.2 DOL Feeder Diagram

Integration with PLC/DCS

In the plant, DOL starters are integrated with the PLC/DCS system:

- Contactor auxiliary contacts feed back a "motor running" status to the PLC.
- Overload relay trip contacts feed a "fault" signal for alarm annunciation on the operator HMI.

- The Start command may originate either from a local push button or remotely from the DCS via an interposing relay, depending on the plant's local/remote control philosophy

4.3 Variable Frequency Drive (VFD)

A VFD (also called a Variable Speed Drive or inverter) controls the speed of an induction motor by varying the frequency and voltage of the power supplied to it. Since the synchronous speed of an induction motor is directly proportional to the supply frequency, reducing the supply frequency reduces the motor speed.

The VFD works in three internal stages:

- Rectifier stage: The incoming AC supply (3-phase, 415V, 50 Hz) is converted to DC by a diode bridge or active rectifier. This DC voltage is stored on a capacitor bank (the DC link).
- DC link: The capacitor bank smooths the rectified DC voltage and provides energy storage to supply the inverter stage.
- Inverter stage: Power transistors (typically IGBTs — Insulated Gate Bipolar Transistors) switch the DC link voltage on and off very rapidly in a pattern (PWM — Pulse Width Modulation) that produces a variable frequency AC output voltage. By controlling the switching pattern, the output frequency and voltage can be set to any desired value.

VFDs are used in the cement plant primarily for large fan drives (ID fan, raw mill fan, cement mill separator) because fan power consumption follows the cube law: reducing fan speed by 20% reduces power consumption by approximately 50%. This makes VFDs one of the most significant energy-saving devices in the plant.

CHAPTER 5

FIELD INSTRUMENTATION

5.1 Introduction to Field Instruments

Field instrumentation refers to all the sensors, detectors and transmitters installed directly on the process equipment in the plant to measure physical variables such as temperature, pressure, level and flow. These devices provide the measurements that the control system needs to monitor and regulate the process. Without accurate, reliable field instrumentation, it would be impossible to maintain the precise conditions required for cement quality and energy efficiency.

5.2 RTD (Resistance Temperature Detector)

An RTD is a temperature sensor that works on the principle that the electrical resistance of a pure metal changes with temperature in a predictable and repeatable way. The most common material used is platinum, giving the Pt100 sensor (100 ohms at 0°C). As temperature increases, the resistance of the platinum element increases linearly. By measuring this resistance, the temperature can be determined accurately.

Parameter	Detail
Operating principle	Change in electrical resistance with temperature
Most common type	Pt100 (Platinum, 100Ω at 0°C)
Temperature range	−200°C to +850°C (industrial use up to ~600°C)
Accuracy	High — better than thermocouple for low/medium temperatures
Output signal	Resistance (Ω), converted to 4–20 mA by a transmitter
Plant applications	Bearing temperature, motor winding temperature, cooling water, compressed air temperature

Table 5.1 RTD characteristics and plant application



Fig. 5.1 4-Wire RTD

5.3 Thermocouple

A thermocouple consists of two wires made of different metals, joined at one end (the measuring junction). When there is a temperature difference between the measuring junction (in the hot process) and the reference junction (at the instrument end), a small voltage is generated — this is the **Seebeck effect**. The magnitude of this voltage depends on the temperature difference and the metals used, allowing temperature to be calculated.



Fig 5.2 Thermocouple Type K

Type	Material pair	Temperature range	Use in cement plant
Type K	Nickel-Chromium / Nickel-Aluminium	−200°C to +1260°C	General purpose — preheater, cooler, kiln shell
Type R	Platinum-Rhodium 13% / Platinum	0°C to +1480°C	Kiln burning zone, high-temperature zones
Type J	Iron / Constantan	−40°C to +750°C	Lower temperature process points
Type T	Copper / Constantan	−200°C to +350°C	Low temperature — cooling water, compressed air

Table 5.2: Thermocouple types used in cement plant

Thermocouples are preferred at very high temperature points such as the kiln inlet, kiln outlet and preheater exit because they can withstand temperatures that would damage an RTD. However, their accuracy is generally lower than that of RTDs, and they require careful handling of the reference junction temperature (cold junction compensation).

5.4 Transmitters and the 4–20 mA Signal Loop

A transmitter is a device that takes the raw output of a sensor (millivolts, ohms, etc.) and converts it into a standardized electrical signal that can be transmitted reliably over long cable distances to the control room. The most widely used standard signal in industrial plants is the 4–20 mA current loop, where 4 mA represents 0% of the measurement range and 20 mA represents 100%.

The reason a current signal is preferred over a voltage signal is that current signals are not affected by the resistance of the cable. In a large plant like a cement factory, cables may run hundreds of metres from field instruments to the control room, and the cable resistance would cause a voltage drop that would introduce errors. A current loop is immune to this effect.

The 4 mA live-zero is an important safety feature: if the cable is broken or the instrument fails completely, the signal drops to 0 mA — which is clearly distinguishable from 4 mA (minimum live reading) and allows the control system to detect and alarm on an instrument failure.

Signal value	Represents
4 mA	0% of measured range (live zero — instrument is working)
20 mA	100% of measured range
0 mA (< 3.8 mA)	Cable break, instrument power failure — fault alarm triggered
>20.5 mA	Instrument over-range or fault condition

Table 5.3: 4–20 mA signal interpretation

5.5 Temperature Transmitter

A temperature transmitter is a dedicated transmitter that connects to an RTD or thermocouple, reads the sensor output, performs linearization and cold-junction compensation (for thermocouples), and outputs a 4–20 mA signal proportional to the measured temperature. Modern smart transmitters also incorporate digital communication (HART protocol) on top of the 4–20 mA signal, allowing remote configuration and diagnostics from the control room.



Fig 5.3 Temperature Transmitter

5.6 Pyrometer

A pyrometer is a non-contact temperature measuring instrument used to measure the temperature of hot surfaces and materials by detecting the infrared radiation emitted by the object. It is widely used in cement plants where temperatures are too high for contact-type sensors.

In the pyro processing section, pyrometers are used to monitor the rotary kiln shell, burner flame, clinker, and clinker cooler temperatures. Continuous temperature monitoring helps maintain stable kiln operation, improves clinker quality, optimizes fuel consumption, and prevents equipment damage due to overheating.



Fig 5.4 Pyrometer

Working Principle:

The pyrometer collects infrared radiation emitted by a hot object through an optical lens. An infrared detector converts this radiation into an electrical signal, which is processed to calculate and display the temperature.

5.7 Switches and Sensors — Proximity Switch, Limit Switch and Load Cells

5.7.1 Proximity Switch

A proximity switch is an electronic sensor that detects the presence or absence of an object without any physical contact between the sensor and the object. This non-contact operation means there are no moving parts in the sensor, resulting in much longer service life compared to mechanical switches and making it suitable for high-speed or high-cycle applications.

The inductive proximity switch, the most common type in this plant, works as follows: An oscillator circuit inside the sensor generates a high-frequency electromagnetic field at its sensing face. When a metallic object enters this field, eddy currents are induced in the metal. These eddy currents consume energy from the oscillator, causing its amplitude to decrease. A trigger circuit detects this amplitude drop and switches the output transistor ON or OFF.



Fig 5.5 Inductive Proximity Switch

Type	Sensing Principle	Detects	Range
Inductive	Eddy current loss in oscillator field	Metallic objects only	2 mm to ~60 mm
Capacitive	Change in capacitance at sensing face	Metals, liquids, plastics, powder	2 mm to ~50 mm
Photoelectric	IR/light beam interruption or reflection	Any opaque object	mm to several metres
Ultrasonic	Time-of-flight of reflected sound wave	Any solid or liquid surface	cm to several metres

Table 5.4: Types of proximity switches

In the cement plant, inductive proximity switches are widely used for:

- Speed monitoring on rotating equipment: A metal tooth or bolt on a rotating shaft passes by the proximity switch face once per revolution, generating a pulse. The frequency of pulses gives the rotational speed. This is used on kiln drives, mill girth gears and conveyor pulleys.
- Zero-speed detection on conveyors: If the conveyor belt stops while the motor is still energized (belt slip or breakage), the proximity switch detects the absence of rotation pulses and triggers a protective trip.
- Position detection: Detecting the presence of a material in a chute or hopper, or confirming that a moving part (such as a gate valve) has reached its open or closed position.

5.7.2 Limit Switch

A limit switch is an electromechanical device that produces an ON or OFF electrical output when a moving part physically contacts its actuator (lever, roller, plunger or rotary arm). Unlike a proximity switch, which operates without contact, a limit switch requires direct mechanical engagement with the target object to switch. The internal contact mechanism uses a snap-action principle, meaning the contacts switch quickly from one state to the other regardless of the speed of the approaching object, preventing arcing and contact bounce.

Every limit switch has at least one pair of contacts: a Normally Open (NO) contact that is open when the actuator is free and closes when the actuator is operated, and a Normally Closed (NC) contact that is closed when the actuator is free and opens when operated.

Application	Type of limit switch	Action on contact
Pull-cord emergency stop on conveyors	NC contact, rope-operated	Pulling rope or rope breaking → opens NC → trips conveyor motor
Stacker/reclaimer rail-end travel limit	NC, lever type	Machine reaches rail end → lever hit → opens NC → reverses or stops travel
Mill inspection door interlock	Positive-opening (safety rated), NC	Door opened → NC opens → mill motor cannot start (or is tripped if running)
Damper position feedback (open/closed)	NO, plunger type	Damper fully open → plunger hit → NO closes → sends position signal to PLC

Belt sway protection	NC, paddle type	Belt sways beyond limit → paddle hit → NC opens → trips conveyor
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Table 5.5: Limit switch applications in the cement plant

3.5.3 Comparison: Proximity Switch vs Limit Switch

Feature	Proximity Switch	Limit Switch
Contact with target	No physical contact	Requires direct physical contact
Moving parts	None (solid state)	Yes (actuator, spring, contacts)
Mechanical wear	None — very long life	Present — contacts and actuator wear
Switching speed	Very fast (kHz range)	Limited by mechanical response
Suitable for	Speed sensing, counting, rotating shaft detection	Position limits, safety interlocks, emergency stops
Robustness	Sensitive to target material type	Rugged — works in dust, oil, moisture
Cost	Higher (electronic)	Lower (electromechanical)

Table 5.6: Comparison of proximity switch and limit switch

5.7.3 Load Cells

A load cell is a transducer that converts a mechanical force (weight or load) into a proportional electrical signal. It is the key sensing element in all weighing applications in the plant.

Working principle: The load cell consists of a precisely machined metal body (called the spring element) on which strain gauges are bonded. When a load is applied, the metal body deforms slightly. The strain gauges, which are thin resistive elements, also deform along with the metal body — stretching in some areas and compressing in others. When a strain gauge stretches, its resistance increases; when it compresses, its resistance decreases.

To maximize sensitivity and compensate for temperature effects, four strain gauges are arranged in a Wheatstone bridge configuration: two gauges experience tension (resistance increases) and two experience compression (resistance decreases) under the applied load. The bridge output is a small voltage signal (typically in the millivolt range) proportional to the applied load. This millivolt signal is then amplified and converted by a weight transmitter into a standard 4–20 mA or digital signal that the PLC can read and use for control.

Location in plant	Purpose
Proportioning weigh feeders	Measure weight of limestone, clay and sand flowing on the weigh feeder belt; PLC adjusts belt speed to maintain the correct flow rate and raw mix ratio
Bag packing machine	Weigh each bag as it is being filled; stop filling when 50 kg is reached
Truck weighbridge at dispatch gate	Weigh loaded trucks to calculate the quantity dispatched
Silo/bin weighing	Monitor the weight of material stored in bins to manage inventory

Table 5.7: Load cell applications in the cement plant

5.8 Final Control Elements — Solenoid Valves and Air Dryers

5.8.1 Overview

In any control loop, the final control element is the device that physically acts on the process in response to a command from the controller. While sensors and transmitters provide the measurement side of a control loop, final control elements provide the actuation side — they change a flow rate, open or close a valve, start or stop a motor, and so on. In a cement plant, the most common final control elements are solenoid valves (for on-off control of pneumatic/hydraulic lines), control valves (for continuous flow regulation) and motor contactors/VFDs (for motor control).

5.8.2 Solenoid Valve

A solenoid valve is an electrically operated valve that uses the magnetic force of an electromagnetic coil (solenoid) to open or close a flow path. When the coil is energized by passing an electric current through it, it creates a magnetic field that pulls a metal plunger into the coil, which in turn opens or closes the valve seat. When the coil is de-energized, a spring returns the plunger to its original position.

Solenoid valves are available in two basic configurations:

- Normally Closed (NC): The valve is closed when de-energized and opens only when the coil is energized. This is the fail-safe configuration for most control applications — if power is lost, the valve closes and blocks flow.
- Normally Open (NO): The valve is open when de-energized and closes when the coil is energized. Used where the fail-safe condition requires the valve to remain open on power loss.

Application in cement plant	Function
Bag filter pulse-jet cleaning	Short bursts of compressed air are blasted through filter bags to dislodge accumulated dust; each solenoid valve controls one row of bags and is pulsed on for ~100 ms every few minutes
Pneumatic actuator control	A solenoid valve controls the supply of instrument air to a pneumatic actuator, which in turn opens or closes a larger process valve or damper
Kiln burner flame management	Solenoid valves control fuel gas or oil supply to the kiln burner as part of the combustion control system
Compressed air tools and utilities	On-off control of compressed air supply to various plant utilities

Table 5.8: Solenoid valve applications in the cement plant

5.8.3 Air Dryers

Compressed air is used extensively throughout the plant for operating pneumatic actuators, cleaning bag filters and supplying instrument air. Before this compressed air can be used in sensitive instruments and actuators, it must be thoroughly dried. This is because compressed air always contains moisture (water vapour from the

atmosphere). When air is compressed, the concentration of water vapour increases, and if not removed, this moisture can cause:

- ☐ Corrosion inside pneumatic valves, cylinders and instrumentation.
- ☐ Freezing in cold weather, blocking small orifices in pneumatic instruments.
- ☐ Contamination of process material if the instrument air comes into contact with the product.
- ☐ Malfunction of solenoid valves and positioners due to water accumulation.

Air dryers remove this moisture from the compressed air before it is distributed to the plant's instrument air network. The most common type used in cement plants is the refrigerated air dryer, which cools the compressed air to a low temperature (typically $+3^{\circ}\text{C}$), causing moisture to condense and be drained out, before the dried air is reheated and distributed. Desiccant-type air dryers (using silica gel or activated alumina) are also used where very low dew-point air is required.

5.9 Pneumatic Systems and Pressure Transmitters

5.9.1 Pneumatic Systems in the Plant

Pneumatic systems use compressed air as the working medium to perform mechanical work — primarily to actuate valves, dampers and gates throughout the plant. The plant maintains a dedicated instrument air system at a regulated pressure (typically 5–7 bar), separately from the utility air used for cleaning and mechanical tools. This instrument air must be clean, dry and at a stable pressure, which is why it passes through filters and air dryers (discussed in Section 3.6) before being distributed.

A typical pneumatic control loop in the plant works as follows: The DCS/PLC controller sends a 4–20 mA (or digital) signal to an I/P converter (current-to-pressure converter), which converts the electrical signal into a proportional pneumatic pressure (typically 3–15 psi). This air pressure signal is applied to a pneumatic actuator, which physically moves the valve or damper to the required position. A position transmitter (LVDT or potentiometer type) on the valve stem provides a position feedback signal back to the controller, completing the control loop.

5.9.2 Pressure Transmitter

A pressure transmitter is an instrument that measures the pressure of a gas or liquid at a point in the process and converts this measurement into a standard electrical signal (4–20 mA) for transmission to the control system.

Working principle: The process pressure is applied to one side of a diaphragm (a thin, flexible metal membrane). The diaphragm deflects by an amount proportional to the applied pressure. This deflection is detected by a sensing element — typically a strain gauge bonded to the diaphragm, or a capacitive element — and converted into an electrical signal. Modern smart pressure transmitters use a silicon capacitive cell: the deflecting diaphragm changes the capacitance between two plates, and this capacitance change is converted electronically into a 4–20 mA output.

Draft (Negative Pressure):

In the kiln and preheater system, a negative pressure (draft) is maintained by the Induced Draft (ID) fan, which pulls hot gases through the system. A pressure transmitter measures this suction and sends a 4–20 mA signal to the PLC to control the ID fan speed via VFD, maintaining the correct draft for stable combustion.

Positive Pressure:

In the compressed air and pneumatic systems of the plant, positive pressure is continuously monitored by pressure transmitters installed on the air distribution lines. The 4–20 mA output signal from these transmitters is read by the PLC to ensure adequate supply pressure is available for solenoid valves, pneumatic actuators and bag filter pulse-jet cleaning systems.



Fig 5.6 Pressure Transmitters showing zero reading during shutdown

Measurement type	Description	Application in plant
Gauge pressure	Pressure relative to atmospheric pressure	Compressed air line pressure, fan discharge pressure
Absolute pressure	Pressure relative to absolute vacuum	Kiln/preheater draft measurement
Differential pressure (ΔP)	Difference in pressure between two points	Bag filter condition monitoring (pressure drop across filter bags), flow measurement in pipes

Table 5.9: Pressure measurement types and applications

5.9.3 Pressure Transmitter Applications in the Cement Plant

- ☐ Compressed air distribution: Monitoring supply pressure to ensure instruments and actuators receive the correct air pressure.
- ☐ Fan pressure monitoring: Measuring the inlet and outlet pressure of ID fans (induced draft) and PA fans (primary air) to calculate the pressure differential they are developing, used for fan performance monitoring and control.
- ☐ Bag filter differential pressure: Measuring the pressure drop across the filter bags in the bag filter. As dust accumulates on the bags, resistance to airflow increases and the differential pressure rises. The control system triggers pulse-jet cleaning cycles based on this reading.
- ☐ Kiln draught: Monitoring the negative pressure (suction) in the kiln and preheater to ensure correct combustion conditions.
- ☐ Silo level estimation: In some installations, the pressure at the bottom of a powder silo is used to estimate the level of material in the silo (pneumatic pressure level measurement).

5.10 Level Switches

Level switches are point-level detection devices employed extensively across the plant to safeguard against silo overflow, hopper blockage, and dry-running of pumps. Unlike level transmitters, which provide continuous level measurement, level switches generate a simple discrete (ON/OFF) signal that is wired to the DCS/PLC for high-level, low-level, and blockage interlocks.

5.10.1 Rotary Paddle Level Switch



Fig 5.7 Rotary Paddle Switch

A small, geared motor continuously rotates a paddle inside the vessel. When bulk material (raw meal, clinker dust, cement) covers the paddle, rotation is obstructed, causing the motor housing to rotate against a spring-loaded microswitch, which trips an alarm/interlock signal. This is the most widely used switch on raw meal and cement silos owing to its reliability in dusty, high-temperature, and coating-prone environments.

5.10.2 Vibrating (Tuning Fork) Level Switch

A forked probe is vibrated at its resonant frequency by a piezoelectric driver. Contact with bulk solid material dampens the vibration amplitude, which is detected electronically and converted to a switch output. These are favoured at transfer chutes and feed hoppers for blockage detection due to minimal moving parts and resistance to material buildup.

5.10.3 Capacitive Level Switch

The probe and vessel wall (or a reference electrode) form a capacitor whose capacitance changes when material (with a different dielectric constant than air) surrounds the probe. This change is detected by the switch circuitry. Capacitive switches are compact, have no moving parts, and are commonly used for high/low-level detection of fine powders in silos.

5.10.4 Diaphragm (Membrane) Level Switch

Bulk material pressing against a flexible diaphragm mechanically actuates an internal microswitch. This is a low-cost, purely mechanical option used at bunkers and hoppers where electronic switches are not justified economically.

5.10.5 Ultrasonic and Radar Level Switches

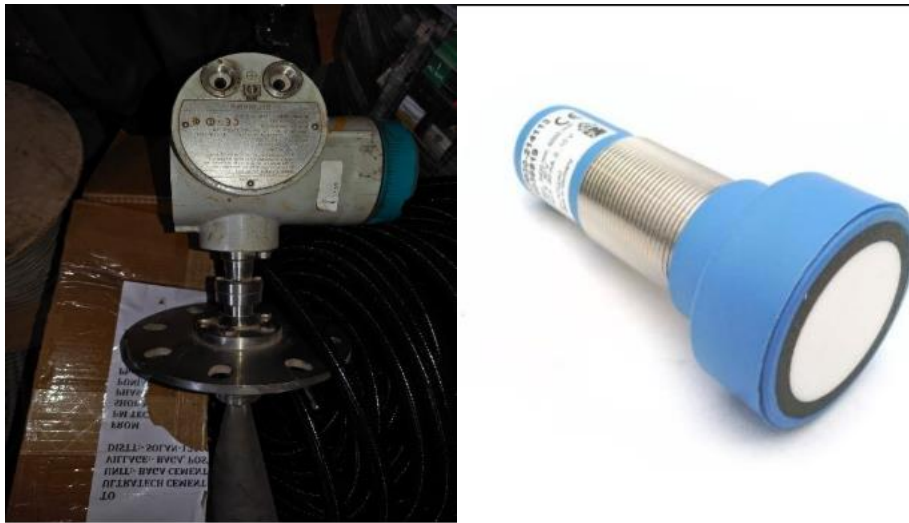


Fig 5.8 Ultrasonic and Radar Level Switch

These are non-contact devices that emit ultrasonic or microwave pulses and detect the reflected signal's time-of-flight to sense the material surface at a fixed switch point. Radar-based switches are preferred in high-dust,

high-temperature silo applications (e.g., clinker silos) where contact-type switches would suffer excessive wear or coating.

5.10.6 Float Switch



Fig 5.9 Float Switch

Used only in liquid applications (cooling water sumps, hydraulic oil reservoirs), a buoyant float rises and falls with liquid level, mechanically or magnetically actuating a reed switch at a set point.

5.10.7 Conductive (Electrode) Level Switch

Applicable to conductive liquids such as water or slurry, this switch detects a change in electrical conductivity between an immersed probe and a reference electrode (or vessel wall) when liquid bridges the gap.

5.11 Analyzers

5.11.1 What is a Gas Analyzer?

A gas analyzer is an instrument that continuously extracts a sample of exhaust gas from the kiln or stack, measures the concentration of specific gases (O_2 , CO , NO_x , SO_2) and transmits this data as a 4–20 mA signal to the local PLC/DCS. In a cement plant it serves two purposes: combustion control (maintaining correct fuel-to-air ratio in the kiln) and environmental compliance (ensuring stack emissions stay within legal limits).



Fig5.10 Gas Analyzer Integrated with Local PLC

5.11.2 Types of Analyzers Used in the Plant

Analyzer	Gas Measured	Principle	Location	Purpose
O ₂ Analyzer	Oxygen (O ₂)	Paramagnetic cell / Zirconia cell	Kiln inlet, preheater exit	Combustion air control — excess air monitoring
CO Analyzer	Carbon monoxide (CO)	NDIR (Non-Dispersive Infrared)	Kiln/preheater exit duct	Incomplete combustion detection, safety interlock
CO ₂ Analyzer	Carbon dioxide (CO ₂)	NDIR	Kiln exit	Calcination and combustion monitoring
NO _x Analyzer	Nitrogen oxides	Chemiluminescence	Stack / chimney	Environmental emission compliance
SO ₂ Analyzer	Sulphur dioxide	UV fluorescence / NDIR	Stack / chimney	Environmental emission compliance
Opacity Monitor	Particulate / dust	Light transmission across stack	Chimney outlet	Bag filter performance monitoring

XRF Analyzer	CaO, SiO ₂ , Al ₂ O ₃ , Fe ₂ O ₃	X-ray fluorescence	Plant laboratory	Raw mix and clinker quality control
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Table 5.10: Types of analyzers used in the cement plant

5.11.3 Working — Process Control (Combustion Control)

The O₂ and CO analyzers work together to maintain correct combustion conditions in the rotary kiln:

- The O₂ analyzer uses a Zirconia cell — a heated ZrO₂ ceramic element that generates a voltage proportional to the difference in O₂ concentration between the process gas and a reference (atmospheric air). This voltage is converted to a 4–20 mA signal representing O₂ percentage.
- The CO analyzer uses the NDIR principle — an infrared light source passes through the gas sample. CO absorbs IR at 4.65 μm; the detector measures the reduction in IR intensity, which is proportional to CO concentration (Beer-Lambert law). Output is a 4–20 mA signal in ppm.
- If O₂ drops below the setpoint → too little combustion air → PLC increases primary air fan speed via VFD.
- If CO rises above the alarm limit → incomplete combustion → PLC reduces kiln feed rate and generates alarm on SCADA.

5.11.4 Working — Environmental Compliance Monitoring

The NO_x and SO₂ analyzers monitor the stack exhaust continuously to ensure the plant meets pollution control norms:

- NO_x analyzer — uses the chemiluminescence principle: NO in the sample reacts with internally generated ozone (O₃), producing excited NO₂ molecules that emit light. The intensity of this emitted light is proportional to the NO_x concentration.
- SO₂ analyzer — uses UV fluorescence: SO₂ molecules absorb UV radiation and re-emit fluorescent light at a longer wavelength. The fluorescence intensity is proportional to SO₂ concentration.
- Opacity monitor — a light beam is sent across the chimney to a receiver. Dust particles reduce light transmission. A rising opacity value indicates bag filter bag failure — PLC generates an alarm on SCADA.
- All emission values (NO_x, SO₂, opacity) are continuously logged by the SCADA historian with timestamps for submission to pollution control authorities.

5.11.5 Integration with Local PLC

The analyzer communicates with the local PLC through analog signals (4–20 mA, one per gas) and digital contact signals (fault, calibration status). The PLC uses these for control actions, alarms and calibration management.

The gas analyzer transmits measured gas concentrations (O₂, CO, NO_x, SO₂) to the local PLC as 4–20 mA analog signals through its analog input cards, which are displayed as live values on the SCADA screen. Based on these inputs, the PLC executes control actions — adjusting primary air fan speed via VFD when O₂ drops below setpoint or reducing kiln feed rate and triggering a SCADA alarm when CO exceeds the high alarm limit. Additionally, the analyzer sends digital contact signals to the PLC for fault indication, low sample flow alarm and calibration mode status, ensuring the PLC suppresses false alarms and holds the last valid reading during the automatic calibration cycle.

CO_HIGH_ALARM bit is set when the CO analog input value exceeds the programmed alarm threshold. When this condition is true AND the kiln is in auto mode, the PLC simultaneously reduces kiln feed and triggers the SCADA alarm.

Signal	Direction	Type	PLC Action
O ₂ measurement	Analyzer → PLC	4–20 mA analog	Adjust primary air fan speed via VFD output
CO measurement	Analyzer → PLC	4–20 mA analog	Reduce kiln feed / generate CO alarm on SCADA
NO _x / SO ₂ value	Analyzer → PLC	4–20 mA analog	Log emission value; generate alarm if limit exceeded
Analyzer fault	Analyzer → PLC	Digital contact (NO)	Freeze last good value; generate instrument fault alarm
Low sample flow	Analyzer → PLC	Digital contact (NO)	Generate blocked probe/filter alarm on SCADA
Calibration mode	Analyzer → PLC	Digital contact (NO)	Suppress alarms; hold last good value during calibration
Calibration start	PLC → Analyzer	Digital relay output	PLC triggers auto-calibration every 24 hours (timer)

Table 5.11: Analyzer-to-PLC signal connections and actions

5.11.6 Auto-Calibration Sequence (PLC Controlled)

Every 24 hours, the PLC triggers an automatic zero-span calibration cycle:

- ☐ PLC sends calibration start → analyzer switches to zero gas (pure N₂) → checks zero reading.
- ☐ Analyzer switches to span gas (known concentration bottle) → checks and adjusts span reading.
- ☐ Analyzer returns to sample mode → sends calibration complete signal to PLC.
- ☐ PLC releases frozen value, re-enables alarms and logs calibration result (pass/fail, timestamp) to SCADA historian.
- ☐ Throughout the calibration cycle, the PLC holds the last valid measurement on the SCADA screen and suppresses all analyzer-linked alarms to prevent false trips.

CHAPTER 6

INDUSTRIAL AUTOMATION AND CONTROL SYSTEMS

6.1 Introduction to Industrial Control Systems

The cement plant uses two complementary types of control systems working together: the PLC (Programmable Logic Controller) for discrete and sequential logic (switching motors on and off, managing interlocks and sequences), and the DCS (Distributed Control System) for continuous process control (regulating kiln temperature, fan speeds, feed rates). Understanding both systems is a core requirement for an instrumentation and control engineer.

6.2 PLC (Programmable Logic Controller)

A PLC is an industrial digital computer designed specifically to control machines and processes in harsh industrial environments. It continuously reads inputs from field devices (push-buttons, limit switches, proximity switches, transmitters), executes the control logic stored in its memory, and updates outputs to field devices (motor contactors, solenoid valves, indicator lamps, drives).

A PLC consists of the following main components:

- CPU (Central Processing Unit): Executes the control program and manages data.
- Input modules (I/O): Digital input cards receive ON/OFF signals from switches and sensors; analog input cards receive 4–20 mA signals from transmitters.
- Output modules: Digital output cards switch contactors and solenoid valves; analog output cards send 4–20 mA setpoint signals to VFDs or valve positioners.
- Power supply: Provides regulated 24 V DC to the PLC modules and field instruments.
- Communication interface: Connects the PLC to the DCS/SCADA network via Modbus, Profibus or Ethernet.



Fig 6.1 PLC Panel

6.2.1 Ladder Logic

PLC programs are written in ladder logic, which represents the control program as a series of horizontal rungs resembling the rungs of a ladder — similar in structure to the relay-based electrical circuits that PLCs were designed to replace. Each rung contains input contacts (conditions that must be true) and output coils (actions to be performed if the conditions are met).

The two basic elements are:

- Normally Open (NO) contact: Passes current (logic 1) only when the associated input is energized (ON). Represented as — | | — in ladder diagrams.
- Normally Closed (NC) contact: Passes current (logic 1) only when the associated input is de-energized (OFF). Represented as — | / | — in ladder diagrams.

A simple motor start/stop rung example:

[START_PB] ————— [/ STOP_PB /] — [/ OVERLOAD /] — (MOTOR_COIL)

This rung means: energize the motor output coil if the start push-button is pressed AND the stop push-button is not pressed AND the overload relay has not tripped.

6.2.2 ON/OFF Control

ON/OFF (or two-position) control is the simplest control action: the output (e.g., a motor or a valve) is either fully on or fully off based on whether the measured variable is above or below a set threshold. For example, a conveyor belt motor is switched ON when material is required and OFF when it is not — there is no intermediate position. This type of control is implemented in PLC ladder logic and is suitable for applications that do not require precise continuous regulation.

6.2.3 Interlock Logic

An interlock is a logical condition in the PLC program that prevents a piece of equipment from starting, or trips it if running, whenever a safety condition is not satisfied. Interlocks are the primary means of protecting both personnel and equipment in the plant.

A practical interlock example observed during training — the mill door safety interlock:

- The ball mill has an inspection door on the side of the mill body. A limit switch is fitted on this door.
- In the PLC program, the mill motor start command is interlocked with the NC contact of this limit switch — the motor can only start if the door limit switch indicates the door is closed (NC contact is made).
- If the door is opened while the mill is running (e.g., for inspection), the limit switch NC contact opens, which immediately trips the mill motor through the PLC output logic.
- This ensures that the mill cannot be started, or continued running, with the inspection door open, protecting anyone who might be near the rotating mill.

A conveyor start sequence interlock is another common example: in a train of multiple conveyors feeding each other, the conveyors must start in reverse order (furthest from feed first, then progressively back to the feed point) to avoid material piling up on a stationary downstream belt. The PLC enforces this sequence through interlock logic — each conveyor's start is interlocked with a time-delay confirmation that the downstream conveyor has already started and reached running speed.

Interlock type	Example	Purpose
Equipment protection and safety interlock	Mill motor trips if mill lube oil pressure drops below minimum	Protect mill bearings from damage
Personnel safety interlock	Mill motor cannot start if inspection door is open	Protect maintenance personnel
Process sequence interlock	Raw mill cannot start if feed conveyor is not running	Prevent process upsets
Environmental interlock	Kiln cannot run if bag filter fan trips (emission protection)	Prevent dust emission

Table 6.1: Types of interlocks in the cement plant

6.3 SCADA/DCS Architecture in the Plant

6.3.1 SCADA Overview

SCADA stands for Supervisory Control and Data Acquisition. It is a software and hardware system that allows plant operators and engineers to monitor and control the entire process from a central control room. The SCADA system collects real-time data from all field instruments and PLCs, displays it on operator workstations (HMI screens), logs historical data for trend analysis, and allows operators to make setpoint changes or start/stop equipment commands from their workstations without having to go to the field.

6.3.2 DCS (Distributed Control System)

Larger cement plants use a DCS in addition to or instead of a simple SCADA system. A DCS is a control system where the control functions are distributed across multiple controllers located throughout the plant (close to the equipment they control), rather than being concentrated in a single central computer. This distributed architecture improves reliability — a fault in one controller affects only the area it controls, not the entire plant — and reduces the distance between the controllers and the field instruments, shortening cable runs.

The DCS architecture observed at the plant can be described as a three-level hierarchy:

- Level 1 — Field level: All sensors, transmitters, actuators, motor starters and drives installed on the process equipment. They provide measurements and accept control commands.
- Level 2 — Control level: PLCs and DCS controllers located in local control panels distributed around the plant. They execute control logic and communicate with field devices through I/O cards.
- Level 3 — Supervisory level: Operator workstations, engineering stations and data historian servers in the central control room. Operators monitor and supervise the process; engineers configure and tune control loops.

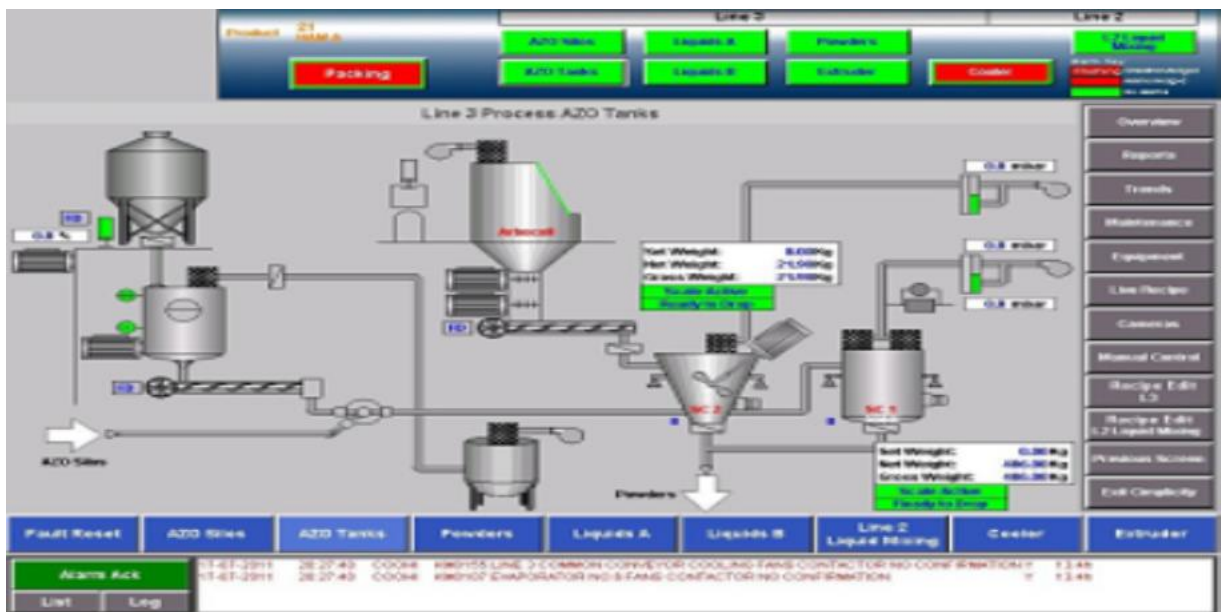


Fig 6.2 HMI inside DCS

6.3.3 HMI (Human-Machine Interface)

The HMI is the interface between the operator and the process — typically a computer screen displaying a graphical representation of the plant (mimic diagram) with live values of temperatures, pressures, flow rates and equipment status overlaid on it. Operators can acknowledge alarms, change setpoints and start/stop equipment through the HMI without leaving the control room. During the training, the control room HMI was observed showing live kiln temperature profiles, preheater stage temperatures, fan speeds, conveyor run statuses and active alarm lists.

SCADA/DCS function	Description
Data acquisition	Reads all field instrument signals continuously (typically every 1 second)
Alarm management	Generates alarms when a process variable goes outside its defined limits; operator must acknowledge each alarm
Historical trending	Stores process data over time; operators can plot trends to diagnose problems or optimise the process
Report generation	Automatic generation of shift/daily production reports
Remote control	Start/stop commands and setpoint changes can be sent to PLCs/drives from operator workstations
Loop control	PID control loops (for kiln temperature, fan speed, etc.) are executed in the DCS controllers

Table 6.2: SCADA/DCS functions observed in the control room

6.4 Communication Protocols — 4-20 mA, HART, Modbus, Profibus and Fiber Backbone

6.4.1 Need for Industrial Communication

A modern cement plant has hundreds of field instruments and dozens of motor drives distributed across a large area. All of these devices must communicate measurements and status information to the central control system, and receive commands in return. The communication standards and protocols used for this purpose are a direct application of the signal and network theory studied in an ECE curriculum.



Fig 6.3 Remote I/O Panel

6.4.2 4–20 mA Current Loop

The 4–20 mA current loop is the most fundamental and widely used analog communication standard in industrial instrumentation. It has been covered in Section 3.4, but its significance as a communication medium is worth emphasising: it is a simple, robust, two-wire communication link between a field transmitter and the control system. The current is the same throughout the loop regardless of cable resistance, making it immune to voltage drops over long cable lengths — a critical property in a plant where cables may run 200–500 metres.

6.4.3 HART Protocol

HART (Highway Addressable Remote Transducer) is a digital communication protocol that is superimposed on the existing 4–20 mA analog signal. It uses frequency-shift keying (FSK) — the same principle as the old telephone modem technology — to transmit a digital signal at 1200 baud on top of the DC 4–20 mA loop current. Because the average value of the FSK signal is zero, it does not affect the 4–20 mA measurement signal.

HART allows engineers to communicate digitally with smart transmitters without disconnecting them from the process. Using a HART communicator (handheld device) or through the DCS, an engineer can remotely read the transmitter's tag, calibration range, diagnostic status, sensor temperature and other parameters while the instrument remains in service, measuring and transmitting the process variable as normal.

6.4.4 Modbus

Modbus is a simple, open, serial communication protocol widely used in industrial automation. It operates on a master-slave architecture: one master device (typically the PLC or DCS) polls multiple slave devices (instruments, drives, I/O modules) in turn, requesting data or sending commands. Modbus is available in two physical variants: Modbus RTU (serial, RS-485) for field devices and Modbus TCP (over Ethernet) for network-level communication between controllers and SCADA servers.

6.4.5 Profibus

Profibus (Process Field Bus) is a faster and more capable industrial fieldbus standard used where many field devices need to communicate with a DCS controller at high speed. Profibus DP (Decentralised Peripherals) is the most common variant in cement plants — it connects distributed I/O modules, smart drives (VFDs) and intelligent field instruments to the DCS controllers over a twisted-pair cable bus. Profibus supports communication speeds up to 12 Mbit/s, making it suitable for time-critical control loops.



Fig 6.4 Profibus (Thick cable) Inside the PLC

Protocol	Type	Speed	Topology	Typical use in plant
4–20 mA	Analog	N/A (continuous)	Point-to-point (1 device per loop)	All process transmitters to PLC/DCS analog input
HART	Digital over analog	1200 baud	Multidrop possible (up to 15 devices)	Smart transmitter diagnostics and configuration
Modbus RTU	Digital serial	Up to 115 kbaud	Multi-drop RS-485 bus	Energy meters, drives, simple I/O to PLC
Profibus DP	Digital serial	Up to 12 Mbit/s	Bus	Distributed I/O, drives, smart instruments to DCS
Ethernet/Modbus TCP	Digital network	100 Mbit/s	Star/ring	PLC-to-SCADA, controller-to-controller communication

Table 6.3: Communication protocols used in the plant

6.4.6 Optical Fiber Backbone

For communication over the long distances within the plant (from the central control room to local panels at the kiln, raw mill, cement mill and packing plant areas, which may be separated by hundreds of metres), the plant uses an optical fiber communication backbone.

Optical fiber is preferred over copper cable for this plant-wide backbone because:

- Immunity to electromagnetic interference (EMI): The large motors, VFDs and high-voltage switchgear in a cement plant generate significant electrical noise. This noise can corrupt data transmitted over copper cables, causing communication errors. Optical fiber transmits data as light, which is completely unaffected by electrical noise.
- Low signal attenuation: Optical fiber can carry signals over much longer distances than copper without significant signal loss, eliminating the need for signal repeaters.

- Electrical isolation: Fiber provides complete electrical isolation between the control room and remote panels, protecting sensitive control equipment from ground faults or surges in the field.
- High bandwidth: Fiber can carry very high data rates, supporting the large volume of data generated by the DCS and SCADA system.

CHAPTER 7

FINDINGS

This chapter presents a personal analysis and reflection on the 34-day industrial training undergone at UltraTech Cement Ltd., Baga Cement Works, Himachal Pradesh. Unlike Other chapters, which focused on technical descriptions of the equipment and systems observed, this chapter answers what was personally learned, what key observations were made on the plant floor, how plant practice compared to classroom theory, and what improvements could be suggested based on the training experience. These findings represent the direct outcome of working inside a large-scale cement manufacturing facility as an Electronics and Communication Engineering student.

7.1 Technical Skills Gained

The most significant outcome of this training was the development of practical technical skills that are difficult to acquire purely from classroom instruction. The following skills were gained or substantially strengthened during the training period:

7.1.1 Identifying Field Instruments by Sight

In a laboratory setting, instruments are neatly labelled and presented in isolation. On the plant floor, they are installed in awkward locations — on pipe bends, at the tops of silos, inside vibrating conveyor structures — often covered in dust and unmarked except for a small tag number. During the training, the ability to identify RTDs (by their thermowell and connection head), thermocouples (by their distinctive ceramic extension cable), pressure transmitters (by their diaphragm housing and impulse tubing), load cells (by their mounting arrangement under weigh feeder structures) and proximity/limit switches (by their mounting brackets and target gaps) purely by physical appearance was developed. This visual recognition skill is essential for any site-based instrumentation work.

7.1.2 Understanding the 4–20 mA Loop in a Real Wiring System

The 4–20 mA current loop was familiar from textbooks as a concept, but the training provided the opportunity to see how it is physically implemented. At the plant, a complete loop was traced from a pressure transmitter on the raw mill inlet duct through the instrument cable, through the marshalling cabinet (where cables are terminated and labelled), through the fiber-optic-connected remote I/O panel, to the analog input card on the DCS controller,

and finally to the live pressure reading displayed on the SCADA screen in the control room. Understanding this complete physical path — including how cable shields are earthed at one end only to prevent ground loops, how terminal blocks are used in marshalling cabinets, and how the DCS scales the 4–20 mA signal into an engineering unit — was a major practical learning that textbooks do not cover in this level of detail.

7.1.3 Reading PLC Ladder Logic on a Live SCADA/HMI Screen

Modern SCADA/DCS systems allow maintenance engineers to view the live status of PLC ladder logic rungs directly on the HMI screen, with contacts highlighted in colour to show their current state. During the training, a simple mill door interlock rung was traced on the HMI: the limit switch input contact, the motor run status coil, and the interlock output were all visible and their live states confirmed against the actual physical equipment status in the field. This made the abstract concept of ladder logic — studied as a programming language in the automation coursework — immediately concrete and meaningful.

7.1.4 Understanding VFDs in Practice

Variable Frequency Drives (VFDs) had been studied in the power electronics coursework in terms of their rectifier-inverter circuit theory. At the plant, it was possible to observe a VFD panel for the ID fan drive and understand how the drive's speed setpoint (received as a 4–20 mA signal from the DCS) is converted into a frequency command that controls the fan speed. The energy saving was made tangible by the plant engineer, who demonstrated that at 80% fan speed the power consumption drops to approximately 51% of the full-speed power — a direct consequence of the cube law of fan power. This practical demonstration made the theoretical energy-saving argument for VFDs far more convincing than a textbook calculation alone.

Skill	Before Training	After Training
Identifying field instruments	Only in lab/textbook	Can identify by sight in plant environment
4–20 mA loop	Understood as circuit concept	Can trace complete physical loop from sensor to SCADA
PLC ladder logic	Understood as programming language	Can read live ladder logic on HMI and match to physical equipment

VFD operation	Understood rectifier-inverter theory	Can relate speed setpoint, output frequency and energy saving in a real drive
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Table 7.1: Summary of technical skills before and after training

7.2 Observations and Key Findings

Beyond individual skill development, the training period produced several broader observations about how a large cement plant functions as a system, and about the role of electronics and communication engineering within it:

7.2.1 Instrumentation Directly Controls Product Quality

One of the most significant observations was how directly the accuracy of field instrumentation affects the quality of the final cement. At the proportioning equipment, load cells on the weighfeeders continuously measure the flow rate of limestone, clay and sand. If a load cell drifts out of calibration by even a small percentage, the raw mix ratio changes, which alters the chemistry of the raw meal fed to the kiln. This, in turn, affects the clinker chemistry and ultimately the strength and setting properties of the finished cement. A small instrumentation error at the start of the process therefore has a cascading effect on product quality at the end. This observation gave a clear and practical understanding of why instrument calibration and maintenance are treated as critical activities in the plant.

7.2.2 Interconnectedness of the System — How One Sensor Failure Affects the Whole Plant

During the training period, a situation was observed where a proximity switch monitoring the speed of a bucket elevator tail pulley gave a false zero-speed signal due to a loose target bolt. The PLC interlock logic interpreted this as a belt stoppage and immediately tripped the elevator motor. Since the elevator was part of the raw mill feed chain, the raw mill also tripped through the process sequence interlock. This single faulty proximity switch (a small, inexpensive device) caused the shutdown of a major section of the plant for approximately one hour while the fault was diagnosed and rectified. This experience clearly demonstrated how deeply interconnected the instrumentation and control system is, and why correct installation, secure mountings and regular inspection of even small field devices is essential.

7.2.3 Coexistence of PLC and DCS in the Same Plant

Before the training, the difference between a PLC and a DCS was understood mainly from definitions. In practice, both systems were observed working together in the same plant on the same process. The PLC handled discrete,

on-off logic: motor start/stop sequences, interlocks, conveyor safety trips and equipment status monitoring. The DCS handled continuous process control: kiln temperature regulation, fan speed control based on draught pressure, and raw mill feed rate control using PID loops. The two systems communicate with each other over the plant network — the DCS can send start commands to the PLC-controlled motors, and the PLC sends equipment run-status signals back to the DCS for display on the SCADA screen. Understanding this division of roles, and how the two systems are integrated, was a key practical insight.

7.2.4 Optical Fiber as the Communication Backbone

The plant covers a very large area — the distance from the control room to the kiln section alone is several hundred metres. During the training, it was observed that all the long-distance communication links between the central control room and the remote I/O panels (at the raw mill, kiln, cement mill and packing plant) use optical fibre cables. When the plant engineers were asked why fiber was used instead of the much cheaper copper cable, the explanation was the combination of the plant's physical size and its severe electromagnetic interference environment: the large VFD panels, motor control centres and high-voltage cabling generate electrical noise that corrupts data transmitted over copper over these distances. Optical fiber, transmitting data as light, is completely immune to this electrical noise. This was a direct practical confirmation of a concept studied in the Optical Fiber Communications coursework.

7.2.5 Gradual Adoption of Wireless Sensing

It was observed that while the plant's primary control and safety instrumentation remains entirely wired (a deliberate choice for reliability and safety), wireless sensing technology is beginning to be used for condition monitoring applications on equipment that is difficult to wire permanently. The stacker, which moves along a rail track, is one such example where wireless vibration monitoring of the travel-wheel gearbox was discussed with the maintenance team as a future improvement. This observation confirmed that wireless IIoT technology is not replacing wired instrumentation in industrial plants but is being added as a supplementary layer for specific applications where wiring is impractical.

7.3 Comparison of Theory and Practice

A recurring theme throughout the training was the difference between how instruments and systems are described in textbooks and how they actually appear and behave in a live industrial environment. The following comparisons summarise the most significant gaps and connections observed

Topic	Theory (Classroom)	Practice (Plant floor)
RTD / Thermocouple	Described as a sensing element with a resistance or voltage output	In the plant, installed inside a thermowell (a protective tube), connected via shielded extension cables, mounted at specific insertion depths into process ducts — aspects not covered in textbooks
4–20 mA signal loop	Taught as a two-wire current loop with a transmitter and receiver	In practice, involves careful cable shielding, earthing only at one end to avoid ground loops, terminal block labelling, and marshalling cabinets with hundreds of loop cables organised by area
PLC ladder logic	Taught as a programming language with contacts and coils	On the HMI screen, live ladder rungs can be watched changing state in real time as equipment starts and stops — immediately connects programming theory to physical machine behaviour
Modbus / Profibus protocols	Studied as communication standards with defined frame formats	Seen as physical RS-485 cables (twisted pair, terminated with a 120-ohm resistor at each end) connecting instruments and drives in a daisy-chain bus topology
VFD (Variable Frequency Drive)	Taught as rectifier-DC link-inverter circuit with PWM output	Observed as a large panel with cooling fans, harmonic filters on the input side, and a motor cable with EMI filtering — the practical installation details that come with the theoretical circuit
Optical fiber communication	Studied as light-based transmission through glass/plastic fiber	Seen as the actual plant-wide communication backbone — multimode fiber in conduits connecting control room to remote I/O panels, with fiber patch panels in each local control panel room

Table 7.2: Theory vs practice comparison from training observations

7.4 Suggestions and Recommendations

Based on the observations made during the training, the following suggestions are offered in the spirit of continuous improvement. These are not criticisms of the existing plant systems, which are well-maintained and effective, but are ideas for future enhancements that align with current trends in industrial automation and IIoT technology:

7.4.1 Wireless Sensors on the Stacker-Reclaimer System

The stacker moves continuously along a rail track in the raw material yard, making permanent wiring of sensors on the travelling machine body expensive and maintenance-intensive (trailing cables or cable reels are needed). It is recommended that the vibration monitoring of the stacker's travel-wheel gearboxes and boom conveyor drive bearings be implemented using battery-powered wireless vibration sensors communicating over LoRaWAN — a long-range, low-power wireless protocol suitable for the large open area of the raw material yard. This would provide continuous bearing health data without the need for trailing cable infrastructure and would enable early detection of bearing wear before it leads to an unplanned breakdown.

7.4.2 Predictive Maintenance for Critical Rotating Equipment

Currently, maintenance of critical rotating equipment such as kiln support roller bearings, mill pinion and girth gear bearings, and large fan bearings is largely time-based (serviced at fixed intervals regardless of actual condition). It is recommended that wireless vibration sensors be added to these critical bearing points, with the sensor data fed to an analytics platform that monitors vibration trends over time. When the trend indicates abnormal vibration (a sign of developing bearing damage), a predictive maintenance alert can be sent to the maintenance team's mobile devices, allowing the bearing to be replaced during a planned shutdown rather than following an unexpected failure. The cost savings from avoiding even one major unplanned kiln shutdown would justify the investment in this system.

7.4.3 Centralised Energy Monitoring Dashboard

The plant has a large number of motor drives — conveyors, fans, mills, pumps — each consuming electrical power continuously. Currently, energy consumption is monitored at the substation level (overall power draw for each section) but not at the individual motor level in real time on a unified dashboard. It is recommended that wireless energy monitoring sensors (clamp-on current transformers with wireless data transmission) be added to the motor control centres of major drives, with all data aggregated on a centralised energy dashboard integrated with the existing SCADA system. This would allow the operations and electrical teams to identify high-

consumption outliers, detect motors running in inefficient load conditions, and quantify the energy saving from VFD-controlled drives compared to fixed-speed operation.

7.4.4 Digital Calibration Records for Field Instruments

During the training, it was observed that calibration records for field instruments (pressure transmitters, temperature transmitters, load cells) are maintained in paper logbooks at the instrument maintenance workshop. This makes it time-consuming to retrieve the calibration history of a specific instrument during troubleshooting. It is recommended that a simple digital instrument management system be implemented — either as a module within the existing SCADA platform or as a standalone database — where each instrument's tag number, calibration date, calibration range, next due date and maintenance history are stored and easily searchable. This would also allow automatic alerts when instruments approach their calibration due date, preventing instruments from running beyond their calibration validity period.

7.5 Relevance of Training to ECE Curriculum

The industrial training at Baga Cement Works directly connected to several core subjects of the B.Tech Electronics and Communication Engineering curriculum, demonstrating that the theory studied in college has direct, practical application in a large industrial setting:

ECE Subject	Connection to Plant Training
Electronic Devices and Circuits	VFD inverter stage uses IGBTs (Insulated Gate Bipolar Transistors) as switching elements — direct application of power semiconductor device theory

Network Analysis	The 4–20 mA current loop and Wheatstone bridge circuit in load cells are practical two-terminal network problems; signal path from field to control room is a real transmission-line application
Signals and Systems	The 4–20 mA signal is a continuous-time analog signal; the DCS samples it at regular intervals (typically 1 second) converting it to a discrete-time digital signal — direct application of sampling theory
Control Systems	PID control loops running in the DCS for kiln temperature and fan speed are real implementations of the closed-loop control systems studied in the Control Systems subject
Optical Fiber Communication	The plant-wide fiber backbone uses multimode optical fiber for data transmission — directly applies fiber types, attenuation, connectors and patch panel concepts from the OFC subject
Computer Networks	Modbus and Profibus industrial protocols follow a master-slave architecture similar to the client-server model studied in networking; Modbus TCP runs over Ethernet using the same TCP/IP stack studied in the networks subject
Electromagnetic Theory	The inductive proximity switch works on eddy-current induction in a metallic target — a direct application of Faraday's law and electromagnetic field concepts
Transducers and Instrumentation	RTD, thermocouple, strain gauge load cell and pressure transmitter — every sensor type studied in the Transducers subject was observed working in a real process environment

Table 7.3: ECE curriculum subjects and their plant training connections

7.6 Summary of Findings

The 45-day industrial training at UltraTech Cement Ltd., Baga Cement Works, was a highly productive and educationally valuable experience from an Electronics and Communication Engineering perspective. The principal findings can be summarised as follows:

- Practical instrumentation skills — P&ID reading, instrument identification, loop tracing, ladder logic interpretation and VFD operation understanding — were developed to a working level that classroom study alone could not provide.
- The critical role of accurate instrumentation in product quality control was directly observed through the load cell-based proportioning system and its downstream effect on clinker chemistry.
- The severe consequence of even a single small instrument failure on overall plant operation was personally witnessed, reinforcing the importance of rigorous maintenance of field devices.
- The coexistence and integration of PLC (discrete logic) and DCS (continuous control) in the same plant, and their communication through standard industrial protocols over an optical fiber backbone, connected directly to ECE communication and networking theory.
- Wireless IIoT sensing technology is being progressively adopted in cement plants as a supplementary monitoring layer, particularly for moving or difficult-to-wire equipment.
- Every major ECE subject studied in the B.Tech curriculum — from electronic devices to optical fiber communication to control systems — has a direct, observable application in the operation of a modern cement plant.

CHAPTER 8

CONCLUSION

The 35-day industrial training at UltraTech Cement Ltd., Baga Cement Works, District Solan, Himachal Pradesh, has been one of the most educationally significant experiences of the B.Tech Electronics and Communication Engineering programme. Working inside one of India's leading cement manufacturing plants provided an exposure to large-scale industrial systems that cannot be replicated in a college laboratory, and the learning gained here will serve as a strong foundation for professional engineering practice.

Summary of Work Done

During the training period, the complete cement manufacturing process was studied and observed from quarrying and crushing through raw material stacking, proportioning, raw grinding, preheating, clinkerization, cooling and cement grinding to the final dispatch from the packing plant. The full range of field instrumentation used to monitor and control this process — including RTDs, thermocouples, pressure transmitters, load cells, proximity switches, limit switches, solenoid valves and gas analyzers — was studied in terms of both working principle and actual application in the plant.

The control systems governing the plant — PLC-based discrete and interlock logic, DCS-based continuous process control loops, and the SCADA operator interface in the central control room — were observed and understood in the context of the actual plant. The communication infrastructure linking field devices to the control room — 4–20 mA current loops, HART protocol, Modbus, Profibus and the optical fiber backbone — was traced physically and connected to the communication theory studied in the ECE curriculum. Motor starting and speed control technologies — DOL starters, star-delta starters, soft starters and Variable Frequency Drives — were observed on large plant motors and their energy-saving significance was appreciated in a practical context. Emerging wireless IIoT sensing technologies (WirelessHART, LoRaWAN) for predictive maintenance were also studied as a modern addition to the plant's instrumentation ecosystem.

Key Takeaways

The following are the most important conclusions drawn from the training:

- Instrumentation is not a peripheral function in a cement plant — it is central to product quality, energy efficiency, equipment safety and environmental compliance. Every stage of the process depends on accurate, reliable measurement and control.
- The interconnectedness of industrial control systems is far greater than classroom study suggests. A single faulty proximity switch can trip a mill section; a calibration error in a load cell can affect cement quality downstream. This reinforces the importance of rigorous maintenance and calibration practices.
- Every subject studied in the ECE curriculum — from electronic devices and network analysis to optical fiber communication and control systems — has a direct, visible application in the operation of a modern cement plant. The plant is, in effect, a large living laboratory of applied electronics and communication engineering.
- The gap between theory and practice is real but bridgeable. Instruments studied as circuit elements in textbooks are, in practice, installed in thermowells, mounted in harsh environments, connected through shielded cables and calibrated periodically. Understanding this gap is the first step to becoming an effective instrumentation engineer.
- Modern industrial plants are progressively integrating wireless IIoT technology for condition monitoring and predictive maintenance, creating new opportunities for ECE engineers who understand both the sensor technology and the wireless communication standards.

Personal Reflection

This training has substantially changed the way the subjects studied in the B.Tech programme are understood. Concepts that previously appeared abstract — PID control loops, Profibus frame formats, optical fiber attenuation, IGBT switching in a VFD — now have a clear, concrete meaning because they have been seen working in a real industrial system. The plant engineers and technical staff at Baga Cement Works were generous in sharing their practical knowledge, and working alongside them gave an appreciation of the professional standards, systematic thinking and attention to detail that industrial engineering requires.

The suggestions offered in Chapter IV — wireless sensors on the stacker, predictive maintenance for critical rotating equipment, a centralized energy monitoring dashboard and digital calibration records — reflect the

application of ECE knowledge to real plant problems observed during the training. These suggestions are offered with the understanding that actual implementation would require detailed feasibility study, cost analysis and management approval; they are presented here as the outcome of applying an engineering student's perspective to the challenges of a working plant.

Future Scope

The cement industry, like all major industries, is undergoing a digital transformation. The integration of IIoT (Industrial Internet of Things), machine learning-based predictive analytics, digital twins (virtual models of physical plant equipment that run in parallel with the actual plant and predict behaviour), and cloud-based SCADA platforms are the directions in which the industry is moving. For an ECE engineer, these developments represent enormous opportunity: the skills of sensor design, signal processing, wireless communication, embedded systems and network engineering are at the core of this transformation. The training at Baga Cement Works has provided both the contextual understanding of the process and the practical instrumentation background that will be essential to contribute meaningfully to this future.

In conclusion, the industrial training at UltraTech Cement Ltd., Baga Cement Works has successfully fulfilled its intended purpose: to bridge the gap between academic theory and industrial practice, to develop practical engineering skills, and to provide a real-world context in which the subjects of the B.Tech ECE programme can be understood and applied. It is hoped that the observations, analyses and suggestions presented in this report reflect both the depth of learning achieved during the training period and a commitment to the responsible, thoughtful practice of engineering.